Data Analyst Technical Assessment

Candidate Instructions

Assessment Components	Task 1: SQL Assessment | Task 2: Strategic Analysis | Task 3: Dashboard Development | Task 4: Presentation Slides
Required Deliverables	One final PDF containing SQL queries as editable text, results, assumptions/comments, strategic analysis answers and dashboard screenshots with the dashboard link, plus a presentation deck.
Scoring	Task 1: 15% | Task 2: 50% | Task 3: 25% | Task 4: 10%
Suggested Tools	PostgreSQL, MySQL, BigQuery Sandbox, SQLite, or any equivalent SQL environment. Any dashboard and presentation tool of your choice.
Submission Link	https://forms.gle/NA5eiMGbm7KLzMiN6

1. Case Background

Delivery - On The Go, a third-generation courier company, runs six delivery hubs in Metro City, grouped into three clusters. In the second week of the period, order volume rose sharply in some hubs, and the forecast says the higher volume will continue. Hub managers are asking for more delivery agents.

Management is not sure hiring is the right answer everywhere and wants a data-backed recommendation before approving any budget. You have 14 days of operational data, the company's cost sheet and a 4-week volume forecast.

2. Datasets

Six Excel files are available to download from the links below, with a CSV copy of every table for easy SQL import (CSV timestamps use the format YYYY-MM-DD HH:MM:SS). The data is clean and complete, but no single file answers a question on its own, so you will need to join them. You may import them into PostgreSQL, BigQuery Sandbox, SQLite, or any other environment you are comfortable using.

Dataset File	What it contains	Excel	CSV
hubs	One row per hub: hub_id, hub_name, cluster	Download	Download
agents	One row per agent: hub, salary type, joining date, active status	Download	Download
orders	One row per order: hub, created time, final status, COD amount, delivery charge	Download	Download
order_logs	One row per status event: order, agent, status, hold reason, event time and attempt number	Download	Download

Dataset File	What it contains	Excel	CSV
cost_sheet	Three tabs: agent_salary (salary and allowances by salary type), costs (in BDT) and rules (company policies)	Download	agent_salary costs rules
volume_forecast	Expected orders per hub for each of the next 4 weeks	Download	Download

3. Important Case Notes

- The period is 13 to 26 October 2026. Week 1 is 13 to 19 October and Week 2 is 20 to 26 October. The forecast covers the next 4 weeks, from 27 October to 23 November.
- Every *Assigned* event is one delivery attempt, followed by one outcome: Delivered, Hold or Returned.
- A held order is attempted again the next day. An order can be returned on any attempt (for example when the customer refuses it), and an order that fails its 3rd attempt is always returned.
- Late Dispatch, Not Attempted and Agent Time Shortage are agent-side hold reasons. All other hold reasons are customer-side.
- The standard workload of an agent is 40 attempts a day, and agents can go above it when a hub is overloaded. The planning target is 85% of the standard workload.
- Hubs operate every calendar day, so each forecast week has 7 operating days. For staffing, count every active agent as available every day.
- A Fixed agent costs the monthly salary ÷ 26 (BDT 650) for every day they work. A Variable agent costs the daily allowance plus the allowance per delivery for every day they work.
- Inactive agents left the company at the end of Week 1.
- All salaries, allowances, costs and company policies are in the three tabs of cost_sheet.xlsx. Import each tab as its own table.

4. Task 1 - SQL Assessment

Submission rule. Place all SQL queries in the final PDF as editable text, not screenshots. Show the result of each query and add comments wherever you make assumptions or want to explain your reasoning.

Q1. Daily Agent Productivity

Build a daily table of each agent's assigned, delivered, hold and returned attempts, and show each hub's average deliveries per agent per day. In your PDF, show the first 20 rows of the daily table and the full hub summary.

Q2. Hub Utilisation in Week 1 and Week 2

Compare each hub's attempts and utilisation in Week 1 and Week 2, and flag the hubs above target in Week 2. State how you define utilisation.

Q3. Hold Reasons

For Week 2, find each hub's total holds, the share that are agent-side, and its top two hold reasons.

Q4. What Happens to Held Orders

For orders held at least once, find the share that were finally delivered, returned or are still on hold, by hub.

Q5. High-Return Agents

Find the agents with more than 30 days of tenure on 26 October 2026 whose return ratio is above 15%.

Q6. Cost per Delivered Order

Calculate each hub's cost per delivered order for Week 1 and Week 2.

Q7. Agents Needed for the Forecast

Calculate required agents and the staffing gap for each of the four forecast weeks.

5. Task 2 - Strategic Analysis

Support every answer with numbers from your analysis and state your assumptions.

Q8. The Hiring Request

Hub managers have asked for 8 new agents to handle the forecast. Would you approve the request? Make your case hub by hub, with costs.

Q9. The Budget

Finance can release BDT 70,000 for next month. It must cover all one-time costs and the salary of any new agents for that month. How would you use it, and what do you expect it to return?

Q10. The Risks

What could make your plan fail, and how would you know within the first two weeks?

6. Task 3 - Dashboard Development

- Build a dashboard that helps the operations team monitor hub performance and decide where to act. The content, design and tool are your choice.
- Whichever tool you use, the dashboard must open for anyone with the link, without a login or an access request. If your tool produces a source file (for example .pbix, .twbx or .xlsx), also share a download link to that file.

7. Task 4 - Presentation Slides

- Present your findings and recommendation to the Head of Operations in a deck of maximum 8 slides.
- Attach it as PDF/PPTX, or share a link that opens for anyone with the link without a login.

8. Submission Format

- Candidates will submit through the Google Form provided by the company. Link: Assessment Solution Submission Form
- The form will collect: name, email address (must match the application email), mobile number, updated CV, and submission files. A Google account is required to upload your files.
- The final PDF must include: all SQL queries in editable text with results, any assumptions/comments, the strategic answers, and screenshots of every dashboard page along with the dashboard link.
- Keep the final PDF clean and professional and label each task clearly.